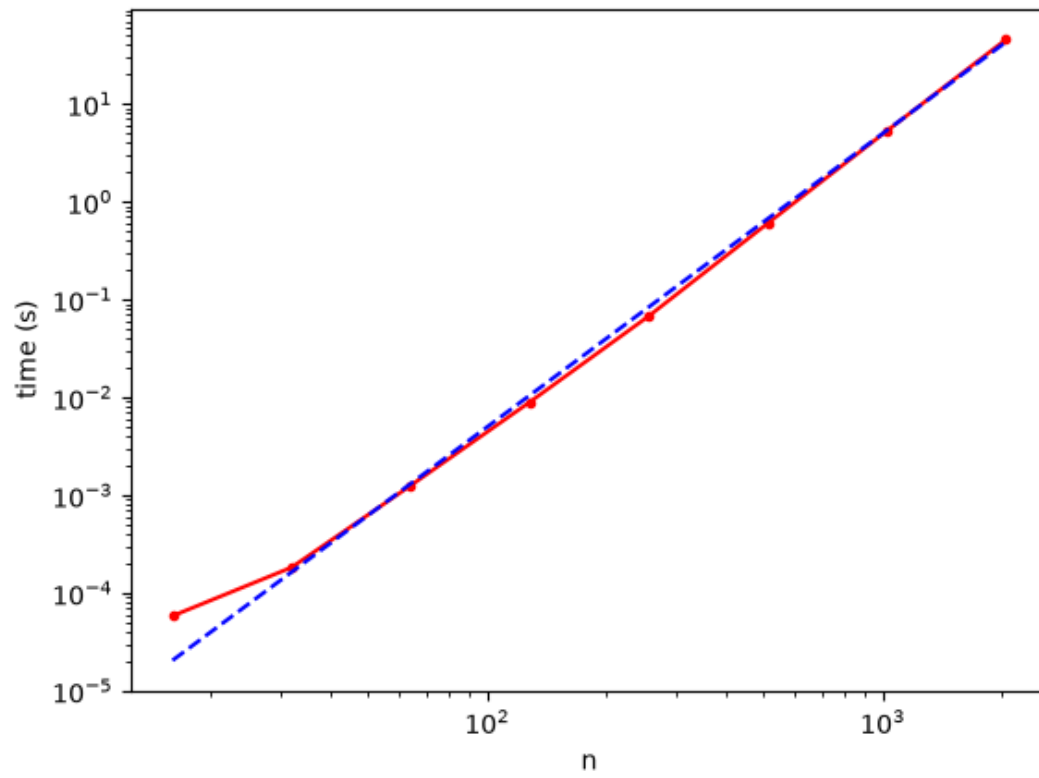


Homework 1

1. Runtime Analysis

(a) $w = 5 \times 10^{-9}$



(b)

```

1 assert a, b, c, n > 0;
2 assert a, b, c ≤ n; // set bounds that a, b, c are at most n

/* get the maximum numbers of each bill for the permutation */
3 max 3 buck ← min(c, n/3);
4 max 2 buck ← min(b, n/2);
5 max 1 buck ← min(a, n);

/* initialize the counter */
6 count ← 0;

7 for the number of max 3 buck do
8   for the number of max 2 buck do
9     if 1. (3 * the number of 3 buck for each iteration) > n then
10      2. (2 * the number of 2 buck for each iteration) > n
11      3. (n - (
12         (3 * the number of 3 buck for each iteration)
13         +
14         (2 * the number of 2 buck for each iteration)
15       ) > n
16     )
17     | do nothing and move on;
18   else
19     | add 1 to count;
20
21 return count

```

Proof:

Let **a** and **b** be the number of 3-buck and 2-buck coins to be combined.

Let **A** be a 2D array that contains all sums of permutations of **a** AND **b** that those sums are always less than or equal to **n**.

So, the number of possible permutations is $a \times b$ and the runtime is $O(n^2)$.

If we add another condition that filters whether the value of each sum in **A** subtracted by **n** does not exceed the maximum number of 1-buck coin provided, we can calculate all the possible permutation and how many the permutations there are at the same time in n^2 times.

Hence, the asymptotic coin choice algorithm can have time complexity of $O(n^2)$.

Runtime Analysis:

1. Variable initialization: the maximum number of coins we can combine and counter $O(1)$

```

int max_three_buck = Math.min(c, n/3);
int max_two_buck = Math.min(b, n/2);
int max_one_buck = Math.min(a, n);
int count = 0;

```

2. Main Loop: $O(n^2)$

- 1) iterate through the maximum numbers of three-buck and two-buck coin ($O(n^2)$)
- 2) eliminate the possible permutations that would go out of bounds from the counter
 - 2-1. if the given number of 3-buck coins worth more than **n**
 - 2-2. if the given number of 2-buck coins worth more than **n**
 - 2-3. if the given number of 3-AND-2-buck coins(permutation) worth more than **n**
 - 2-4. From the permutation without 2-1,2,3., if the left-over money is greater than the maximum number of 1-buck coin.
 - 2-5. Otherwise, put it in the counter.
- 3) Return the number of permutations ($O(1)$)

```

for (int first = max_three_buck; first >= 0; first--){
    for (int second = max_two_buck; second >= 0; second--){
        if(3*first > n || 2 * second > n || (3*first)+(2*second) > n || (n-(2*second+3*first)) >
        max_one_buck){
            continue;
        } else {
            count++;
        }
    }
}
}

```

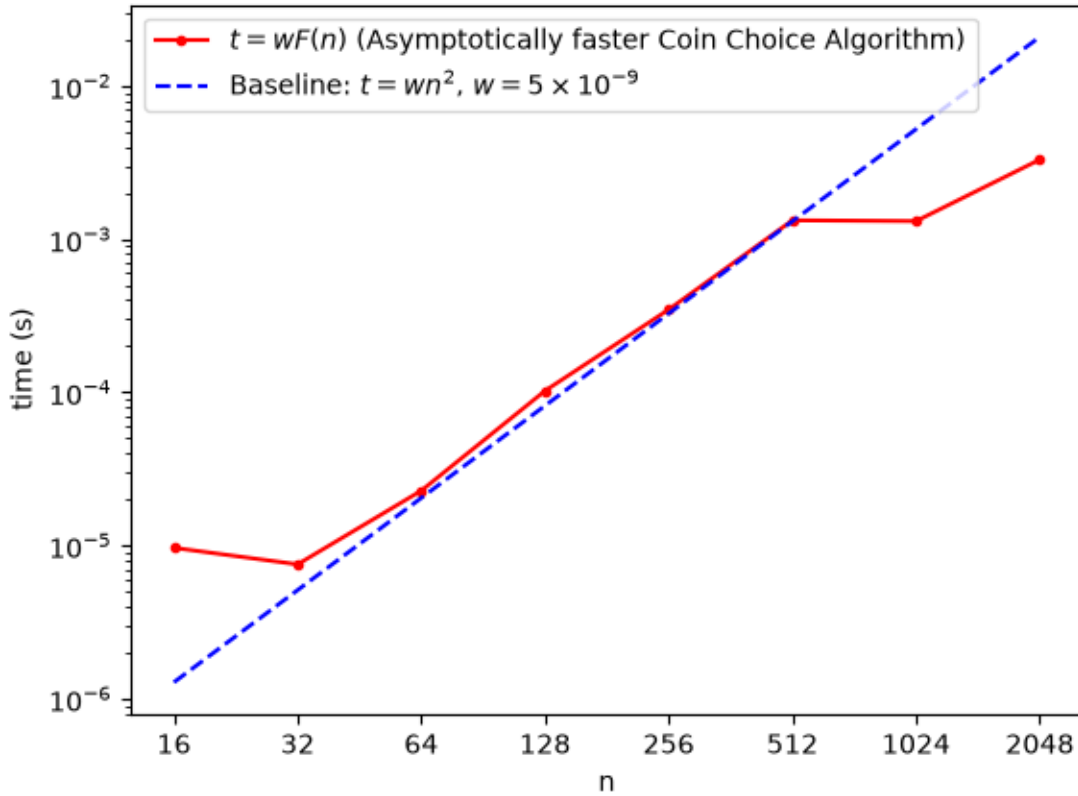
3. Return result: $O(1)$

```
return count;
```

Complete Runtime = $O(1) + O(n^2) + O(1) = O(n^2)$

Therefore, the runtime of this asymptotically faster algorithm for this problem is $O(n^2)$.

(c)



Initially, w value was 5×10^{-8} as default. For part (a), I changed the exponent to -9 , which fits to the original algorithm with time complexity of $O(n^3)$. For the part (c), I did not change w value and still able to fit my asymptotically faster algorithm with time complexity of $O(n^2)$. Because w value is just an adjustment parameter to make the comparison between the baseline $t = wn^2$ and my algorithm human-friendly and my hardware setup was not changed, I did not have to change w value from part (a). One problem I encountered was the data extraction. Since I wrote the algorithm in Java and the original algorithm is in python, I had to come up with an idea that I can put into the plot in python (I don't know how to plot in Java so I had to extract data that can be read cross-linguistically.)

2. Running Sum

Context

- Given arrays A and B (B with n elements),

$$A[i] = B[i] + B[i+1] + B[i+2] + \dots + B[i+r-1]$$

for all $i : 0 \leq i \leq n-r$

- 0-based indexing; $i+r-1$ for upper bound.
- (a) The length of array A is $n-r+1$. Because if the bound for i is $0 \leq i \leq n-r$, since it is inclusive and 0-based indexing, we need to add 1 to get the final array length and not losing the last element of the array A .
- (b) 1st box: 0
 2nd box: `A.length`
 3rd box: `i`
 4th box: `i+r`
 5th box: `A[i] + B[j]`
- (c) 1st box: 1
 2nd box: `A.length`
 3rd box: `A[i-1] - B[i-1] + B[i+r-1]`
- (d) Proof by Induction
 Prove $A_i = \sum_{k=i}^{i+r-1} B_k$
- Base case: $i=0, r=1$

$$A_0 = \sum_{k=0}^{0+1} B_k = \sum_{k=0}^1 B_k = B_0 + B_1$$

it holds!

2. Inductive Hypothesis:

If $A_i = \sum_{k=i}^{i+r-1} B_k$ holds, then $A_i = \sum_{k=i}^{i+r-1} B_{k+1}$ holds.

$$\begin{aligned} A_i &= B_i + B_{i+1} + \dots + B_{i+r} \\ A_{i+1} &= B_{i+1} + B_{i+2} + \dots + B_{i+r} \\ A_{i+1} - A_i &= -B_i + B_{i+r} \\ A_{i+1} &= A_i - B_i + B_{i+r} \end{aligned}$$

The algorithm has a `for` loop of $O(n)$ time complexity that has the same equation as A_{i+1} . $\square$